

Online Food Delivery Analysis

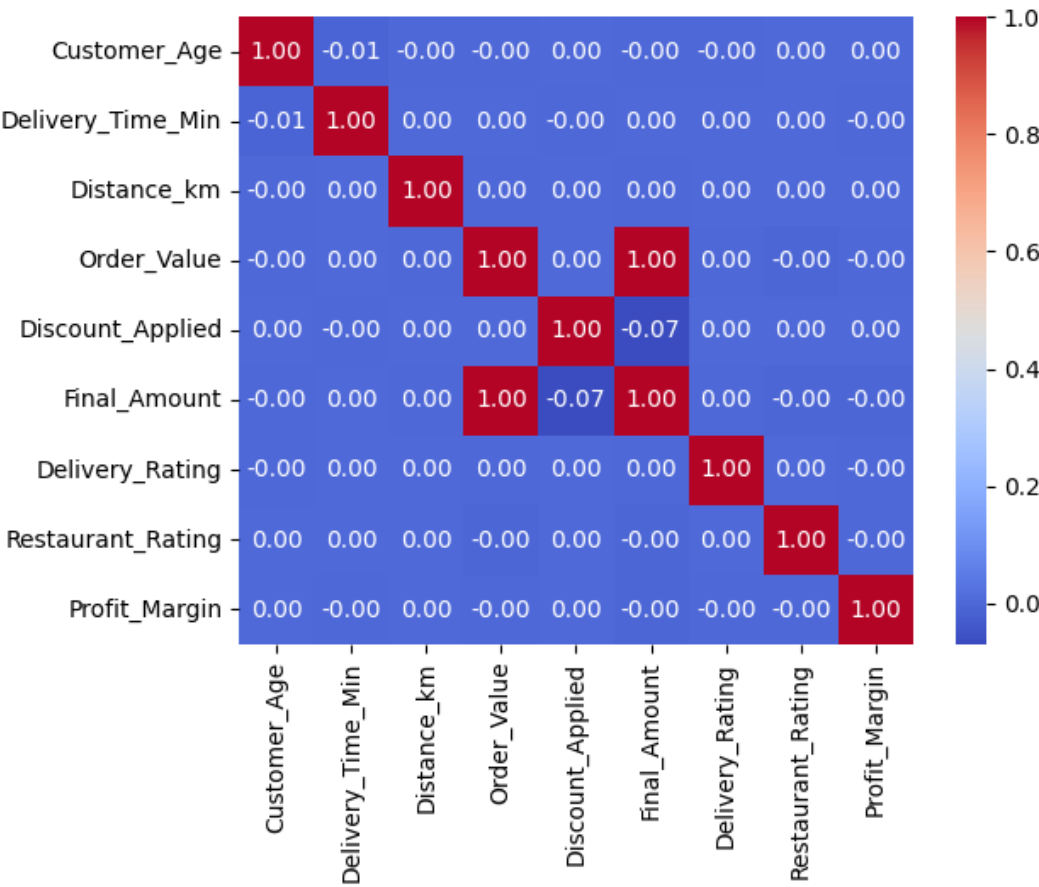
Exploratory Data Analysis Report

Data | Charts | Insights

Cleaned dataset: 98,986 orders

Correlation Analysis (numeric features)

Chart

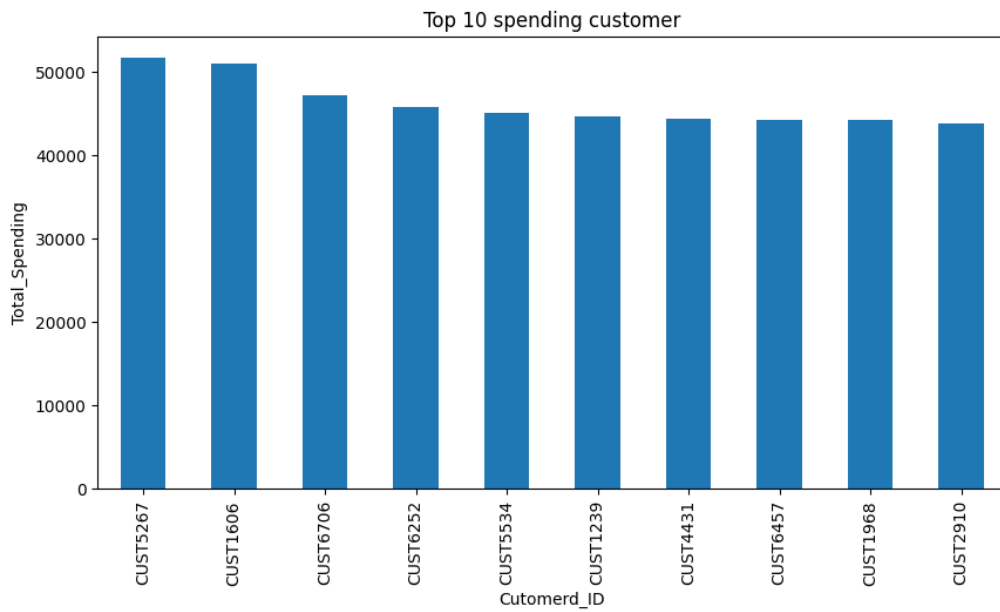


1. Top 10 spending customer

Data

```
Customer_ID
CUST5267    51677.0
CUST1606    50971.0
CUST6706    47239.0
CUST6252    45825.0
CUST5534    45068.0
CUST1239    44733.0
CUST4431    44335.0
CUST6457    44201.0
CUST1968    44181.0
CUST2910    43755.0
Name: Final_Amount, dtype: float64
```

Chart



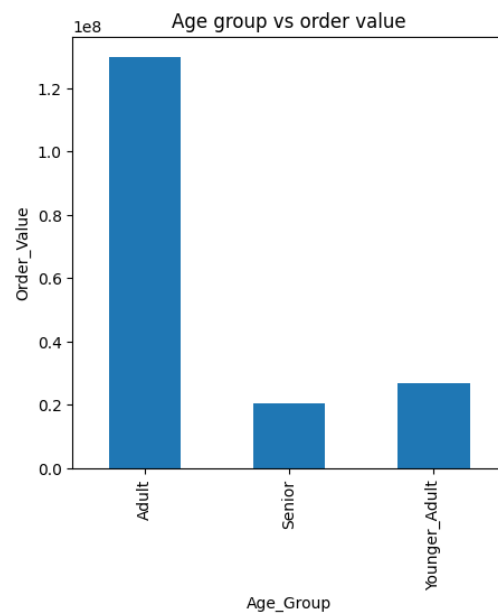
Insight: A few customers contribute significantly to total revenue compare to other customers

2. Analyze age group vs order value

Data

```
Age_Group
Adult      129839205.0
Senior     20322759.0
Younger_Adult 26735783.0
Name: Order_Value, dtype: float64
```

Chart



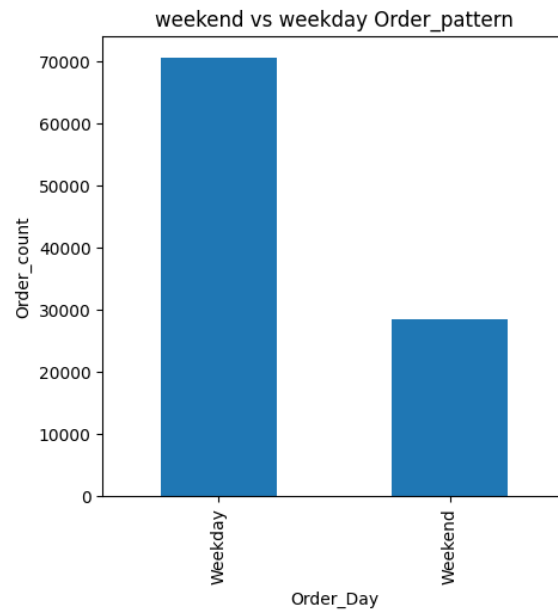
Insight: Most revenue comes by adult compare to senior and younger_adult

3. Weekend vs weekday order patterns

Data

```
Order_Day  
Weekday    70618  
Weekend     28368  
Name: count, dtype: int64
```

Chart



Insight: Most orders happen on weekdays 70,618 orders, compared with 28,368 on weekends.

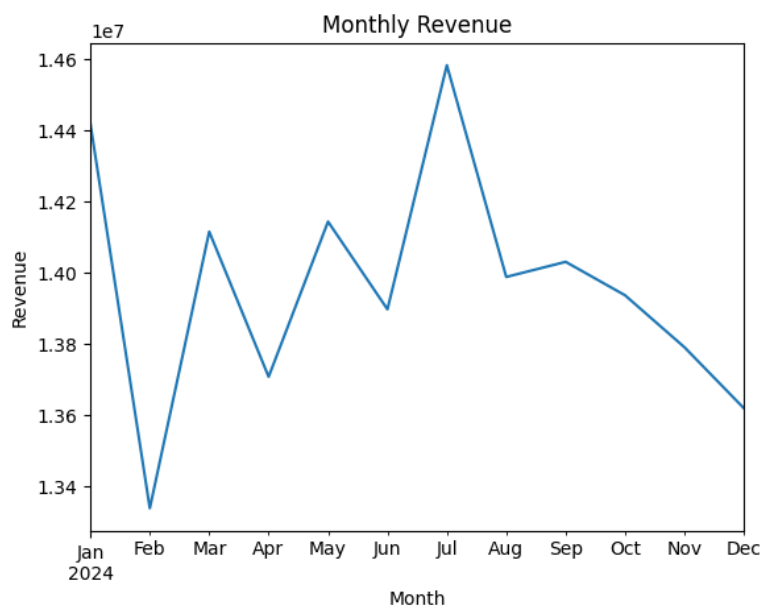
4. Monthly revenue

Data

Order_Date	
2024-01	14416252.0
2024-02	13337626.0
2024-03	14114118.0
2024-04	13706742.0
2024-05	14142429.0
2024-06	13896071.0
2024-07	14581373.0
2024-08	13986979.0
2024-09	14029400.0
2024-10	13935442.0
2024-11	13789582.0
2024-12	13618394.0

Freq: M, Name: Final_Amount, dtype: float64

Chart



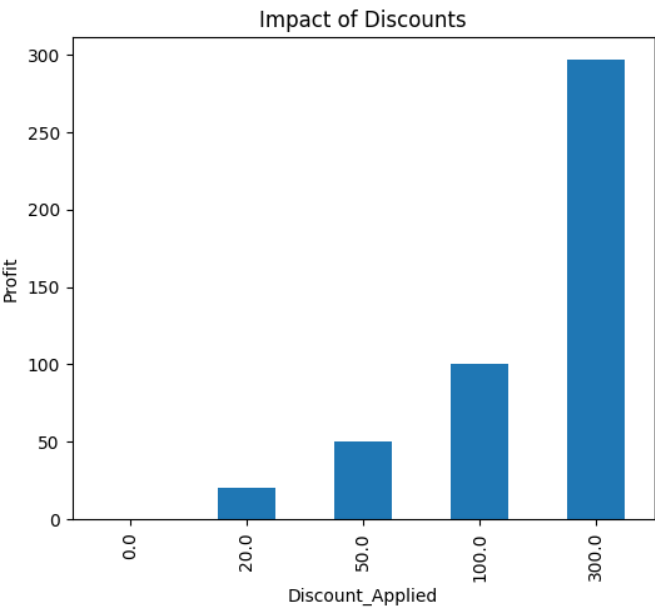
Insight: July have the highest revenue of (14581373) and february has the lowest (13337626)

5. Impact of Discounts on Profit

Data

```
Discount_Applied
0.0      0.000000
20.0     20.000000
50.0     50.000000
100.0    100.000000
300.0    296.413267
Name: Price_Reduction, dtype: float64
```

Chart



Insight: Higher discounts directly increase the price reduction.

6. High-revenue cities and cuisines

Data

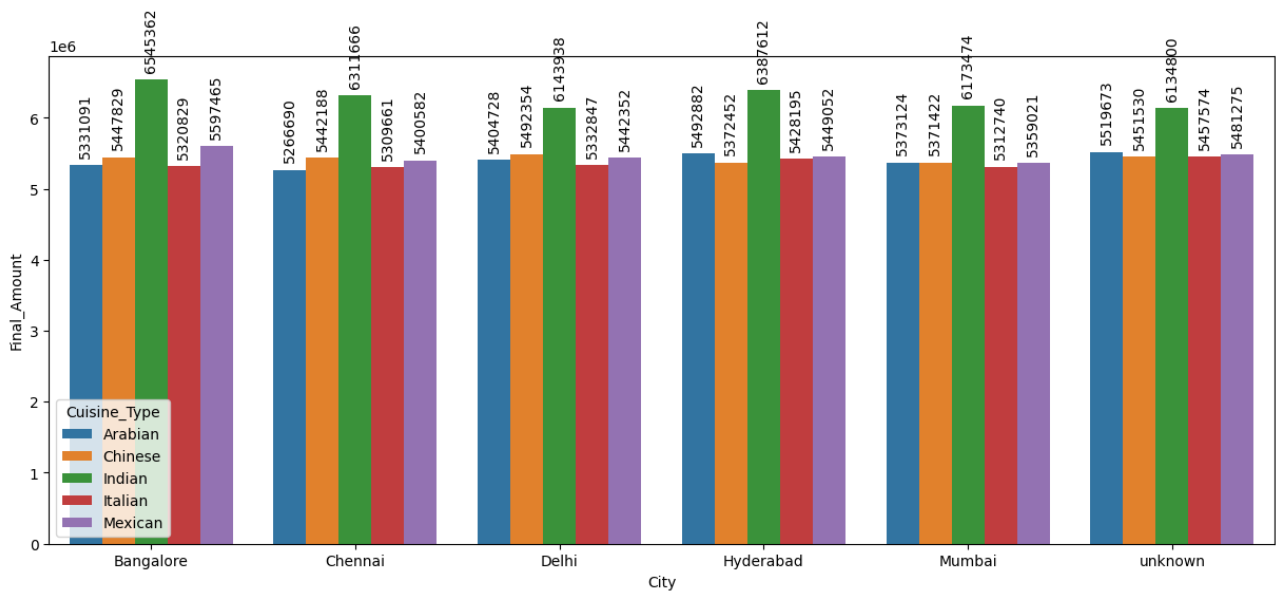
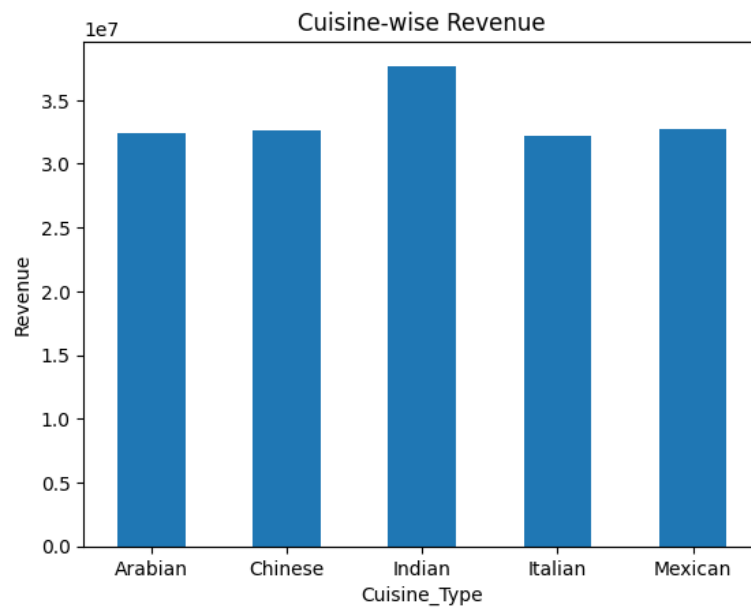
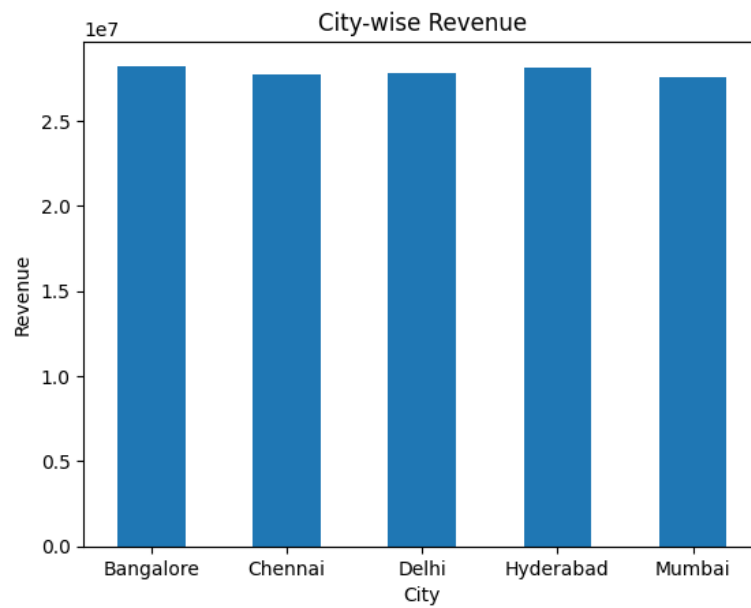
```
Cuisine_Type
Indian      22148
Arabian     19286
Chinese     19219
Mexican     19217
Italian     19116
Name: count, dtype: int64
```

	City	Cuisine_Type	Final_Amount
0	Bangalore	Arabian	5331091.0
1	Bangalore	Chinese	5447829.0
2	Bangalore	Indian	6545362.0
3	Bangalore	Italian	5320829.0
4	Bangalore	Mexican	5597465.0
5	Chennai	Arabian	5266690.0
6	Chennai	Chinese	5442188.0
7	Chennai	Indian	6311666.0
8	Chennai	Italian	5309661.0
9	Chennai	Mexican	5400582.0
10	Delhi	Arabian	5404728.0
11	Delhi	Chinese	5492354.0
12	Delhi	Indian	6143938.0
13	Delhi	Italian	5332847.0
14	Delhi	Mexican	5442352.0
15	Hyderabad	Arabian	5492882.0
16	Hyderabad	Chinese	5372452.0
17	Hyderabad	Indian	6387612.0
18	Hyderabad	Italian	5428195.0
19	Hyderabad	Mexican	5449052.0
20	Mumbai	Arabian	5373124.0
21	Mumbai	Chinese	5371422.0
22	Mumbai	Indian	6173474.0
23	Mumbai	Italian	5312740.0
24	Mumbai	Mexican	5359021.0
25	unknown	Arabian	5519673.0
26	unknown	Chinese	5451530.0
27	unknown	Indian	6134800.0
28	unknown	Italian	5457574.0
29	unknown	Mexican	5481275.0

```
City
Bangalore    28242576.0
Chennai      27730787.0
Delhi        27816219.0
Hyderabad    28130193.0
Mumbai       27589781.0
Name: Final_Amount, dtype: float64
```

```
Cuisine_Type
Arabian      32388188.0
Chinese      32577775.0
Indian       37696852.0
Italian      32161846.0
Mexican      32729747.0
Name: Final_Amount, dtype: float64
```

Charts



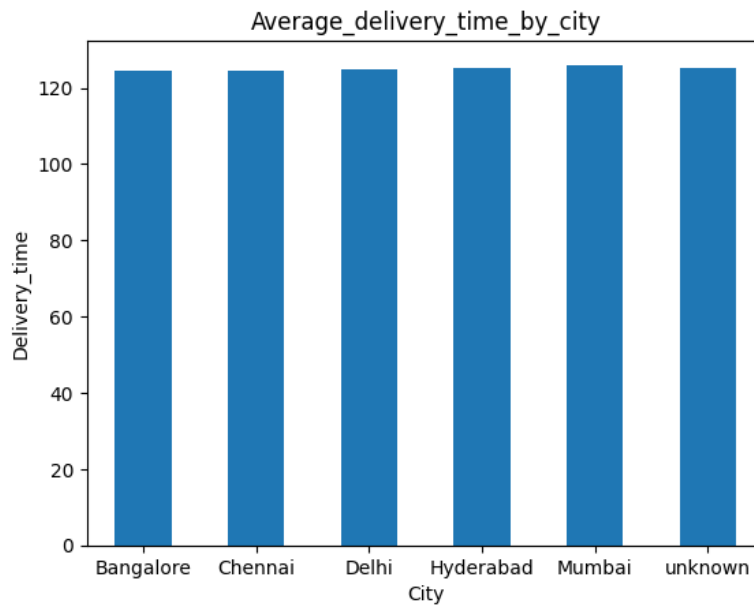
Insight: Bangalore has the highest city revenue and Indian cuisine generates the highest revenue.

7. Average delivery time by city

Data

```
City
Bangalore    124.341402
Chennai      124.584049
Delhi        124.906604
Hyderabad    125.045234
Mumbai       126.039458
unknown      125.051957
Name: Delivery_Time_Min, dtype: float64
```

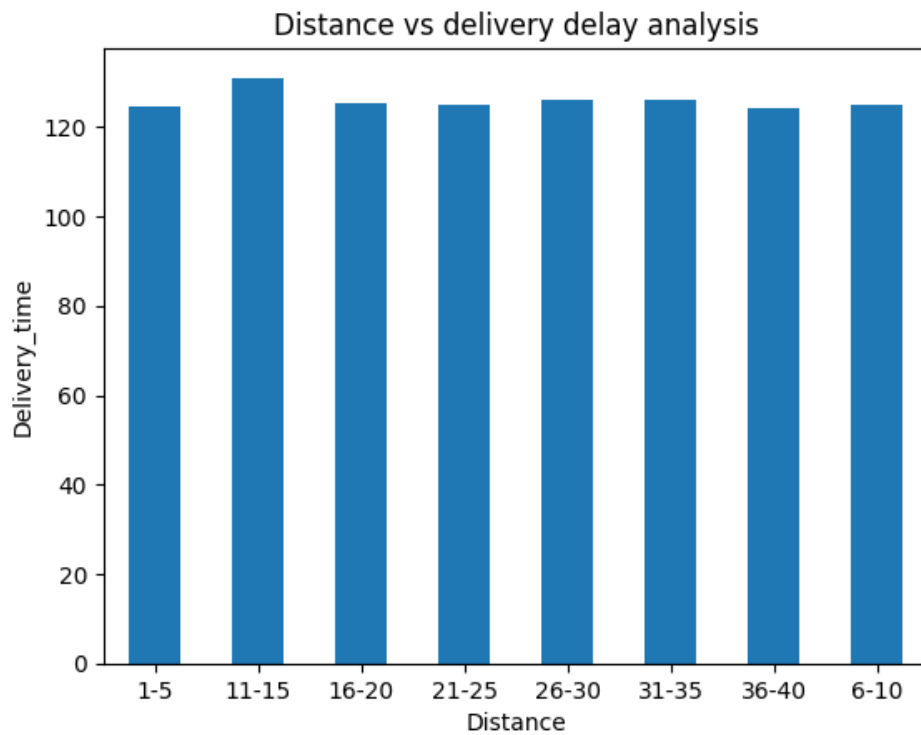
Chart



Insight: Average delivery time is very similar across cities, around 124–126 minutes. Mumbai has the highest average at 126.04 minutes

8. Distance vs delivery delay analysis

Chart



Insight: There is no correlation between distance and delivery_time

9. Delivery rating vs delivery time

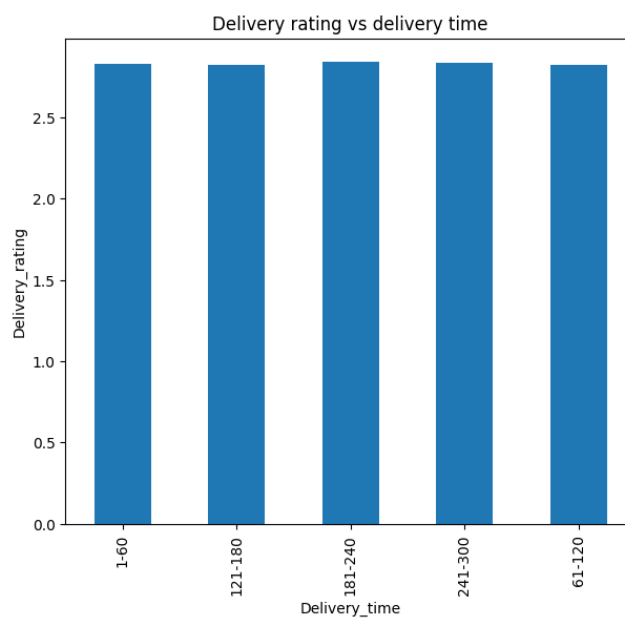
Data

```
group_delivery_time
1-60      2.825605
121-180   2.820341
181-240   2.843557
241-300   2.834704
61-120    2.823809
Name: Delivery_Rating, dtype: float64
```

```
Text(0, 0.5, 'Delivery_rating')
```

```
Text(0, 0.5, 'Delivery_rating')
```

Charts



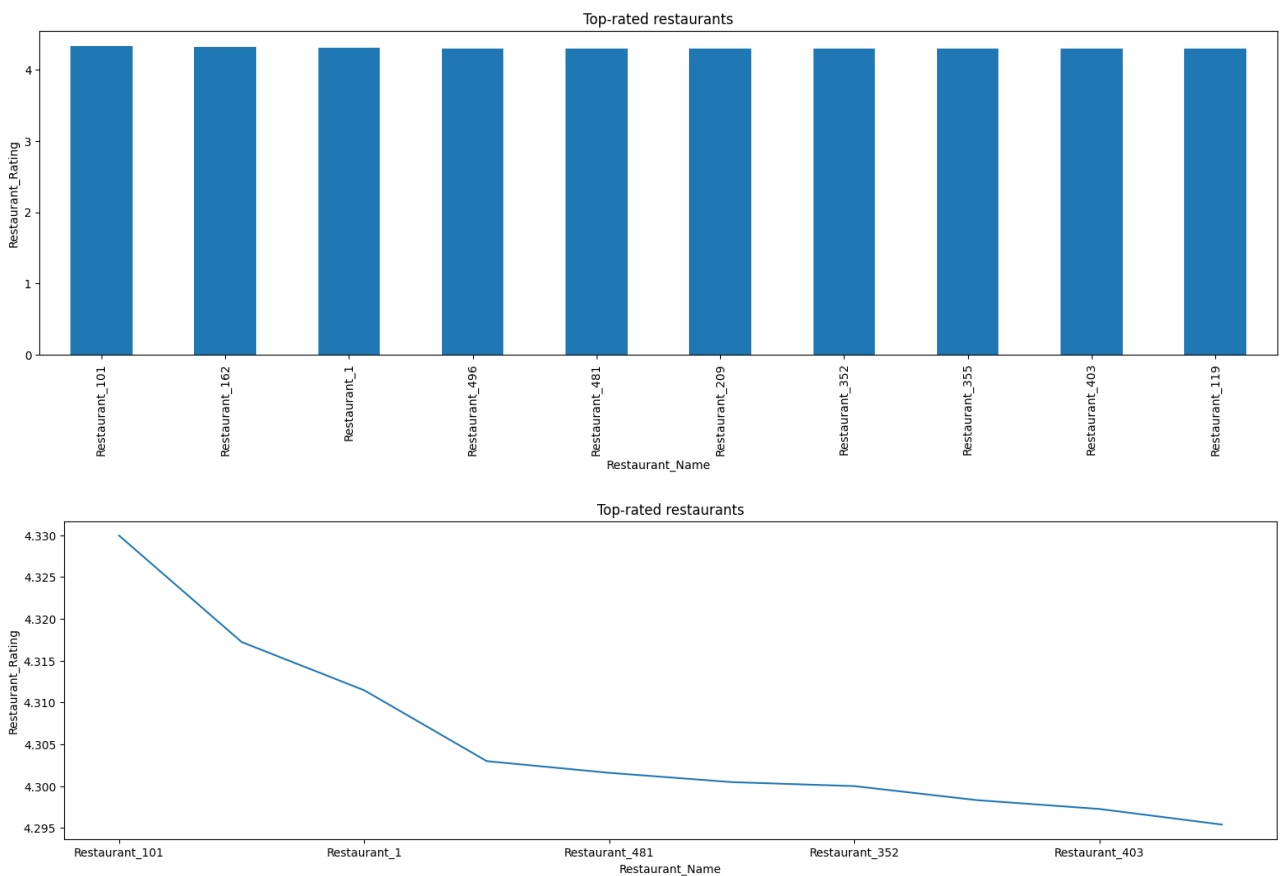
Insight: Delivery ratings are quite similar across delivery-time groups

10. Top-rated restaurants

Data

```
Restaurant_Name
Restaurant_101    4.329944
Restaurant_162    4.317222
Restaurant_1       4.311458
Restaurant_496    4.302985
Restaurant_481    4.301587
Restaurant_209    4.300478
Restaurant_352    4.300000
Restaurant_355    4.298324
Restaurant_403    4.297268
Restaurant_119    4.295402
Name: Restaurant_Rating, dtype: float64
```

Charts



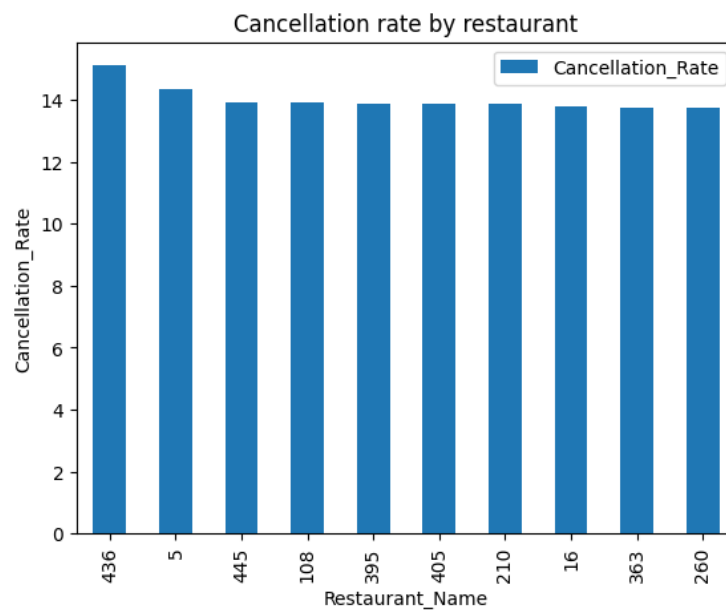
Insight: Restaurant_101 has the highest average restaurant rating at about 4.33/5.

11. Cancellation rate by restaurant

Data

	Restaurant_Name	Cancellation_Rate
436	Restaurant_492	15.104167
5	Restaurant_103	14.347826
445	Restaurant_50	13.917526
108	Restaurant_197	13.903743
395	Restaurant_455	13.888889
405	Restaurant_464	13.888889
210	Restaurant_289	13.875598
16	Restaurant_113	13.793103
363	Restaurant_426	13.744076
260	Restaurant_333	13.725490

Chart



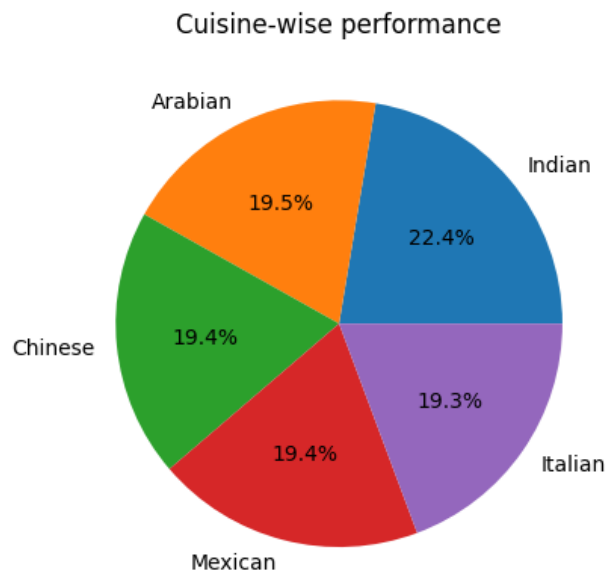
Insight: Restaurant_492 has the highest cancellation rate among the restaurants analyzed, at about 15.10%.

12. Cuisine-wise performance

Data

```
Cuisine_Type
Indian      22148
Arabian     19286
Chinese     19219
Mexican     19217
Italian     19116
Name: count, dtype: int64
```

Chart



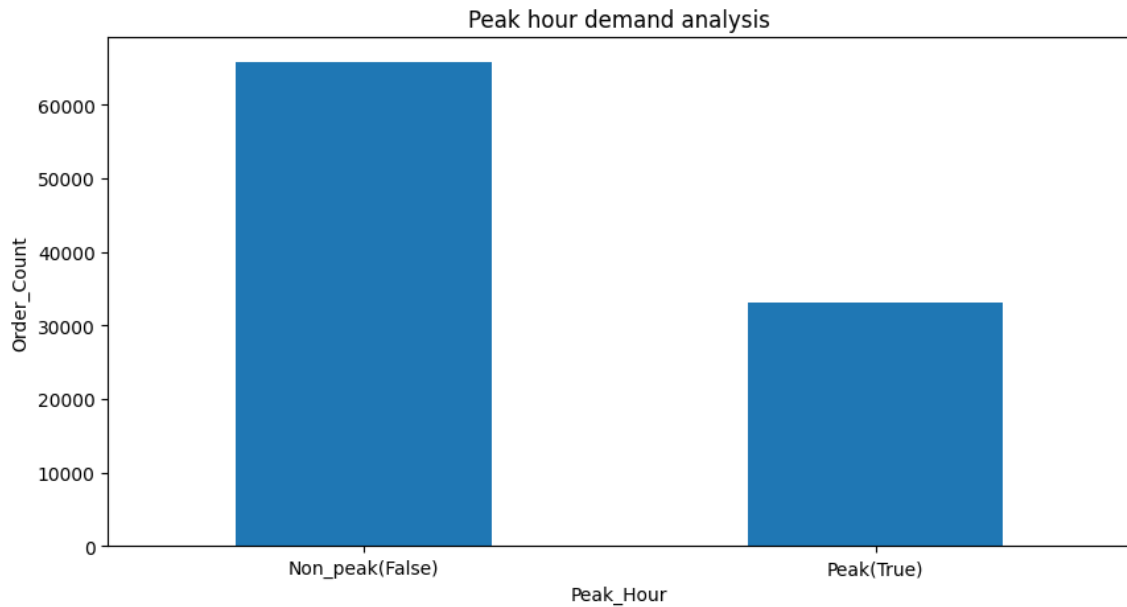
Insight: Indian cuisine performed the best in terms of revenue.

13. Peak hour demand analysis

Data

```
Peak_Hour
False    65845
True     33141
dtype: int64
```

Chart



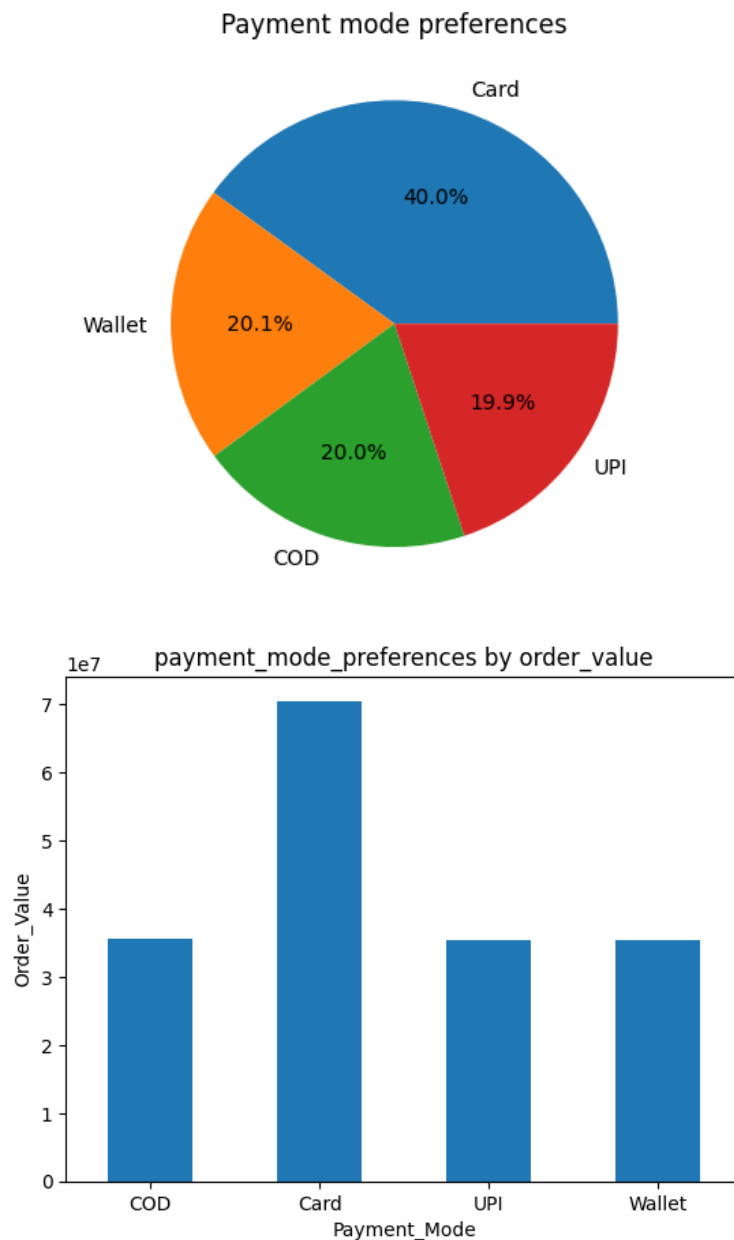
Insight: Non-peak hours have higher order than peak hours.

14. Payment mode preferences

Data

```
Payment_Mode
Card      39603
Wallet    19885
COD       19767
UPI       19731
Name: count, dtype: int64
```

Charts



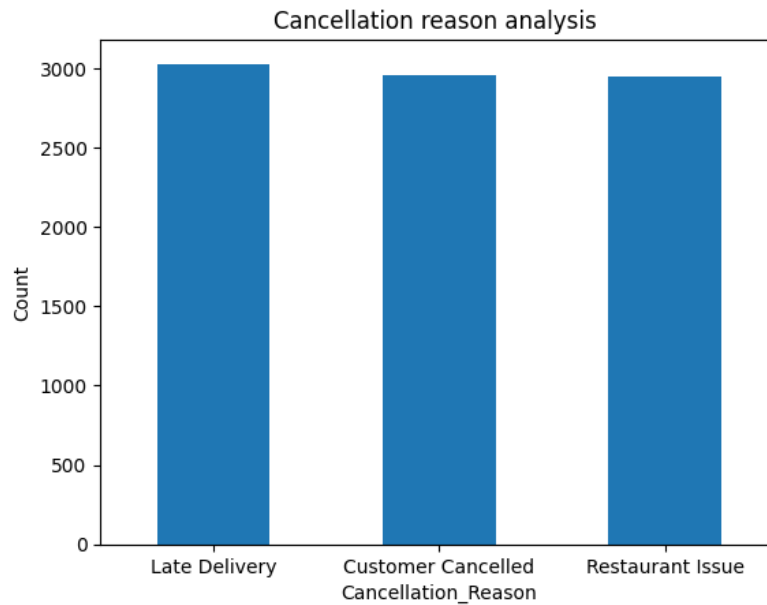
Insight: Card is the most-used payment method with 39,603 orders, followed by Wallet, COD and UPI.

15. Cancellation reason analysis

Data

Cancellation_Reason	
Late Delivery	3032
Customer Cancelled	2963
Restaurant Issue	2949

Chart



Insight: The main cancellation categories are Late Delivery (3,032), Customer Cancelled (2,963) and Restaurant Issue (2,949).

Overall Insights

Insight: From my analysis, I found that adults generate the highest revenue, weekdays have more orders than weekends, and Bangalore has the highest city revenue. Indian cuisine generates the highest revenue. I also found that Restaurant_492 has the highest cancellation rate, while Restaurant_101 has the highest average rating. Card is the most preferred payment method, and late delivery is the most common cancellation reason.